FILM PRODUCTION MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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in partial fulfilment of the End Semester Examination for the course

CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

October, 2026

RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled “Film Production Management System” is the bonafide work of MENAHA R (240701308), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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DECLARATION

I, MENAHA R (240701308), hereby declare that the project report titled “Film Production Management System”, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date:

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ACKNOWLEDGEMENT

I express my sincere gratitude to our Chairman Thiru. S. Meganathan, Chairperson Dr. (Mrs.) Thangam Meganathan and Vice-Chairman Mr. Abhay Meganathan S for providing the infrastructure and facilities that made this work possible.

I am grateful to our Principal, Dr. S. N. Murugesan, Rajalakshmi Engineering College, for the constant encouragement extended to students to take up industry-relevant learning.

I thank Dr. Manoranjini J, Head of the Department, Department of Computer Science and Engineering, for the guidance and support provided throughout the course.

I owe my sincere thanks to my course faculty, Mrs. Dr. J. Jinu Sophia, Assistant Professor, for the valuable guidance, reviews and feedback that shaped this application and report.

I also acknowledge Zoho Corporation for providing access to the Zoho Creator platform, its trial edition and its learning resources, and I thank my family and friends for their unwavering support.

ABSTRACT

A film is made by a temporary team that has to bring together a story, a budget, actors, technicians, locations, equipment and a day-by-day shooting plan. In many production houses these details are still kept in separate spreadsheets, paper call sheets and chat groups, so the producer rarely has one reliable view of what is scheduled, who is available, which equipment is in good condition and how much of the budget has been spent. The Film Production Management System is a low-code application built on Zoho Creator that brings these activities into one hosted application. Zoho Creator was chosen because forms, reports, pages, Deluge scripting, workflows, role-based permissions and a mobile app are available on one platform, which allowed a single student developer to build a complete system within a few weeks. Film Master, Cast and Crew Master, Location Master, Equipment Master and Scene Master forms hold the master data, and lookup fields link them to the Shooting Schedule, Equipment Allocation, Shooting Attendance, Daily Shooting Report and Production Completion forms. A Blueprint named Movie Production guides each film through its production stages, and an approval process reviews every equipment allocation request. Deluge functions calculate the remaining budget and the number of shooting days, check a shooting schedule, classify a movie and validate its rating. A home page dashboard shows the number of requests, the total budget, the number of members and the active projects, together with an equipment chart, a director search and a cast and crew board, while a pivot report summarises equipment condition and rental cost and a bubble chart monitors crew attendance as present, late, half day or absent. A role hierarchy of CEO, producer, director, production manager and crew member, together with permission sets such as Admin, Producer, Director, Cinematographer, Crew Member and Post-Production Manager, controls who can see and change each record, and a portal permission lets actors and crew members view film details, schedules and assigned tasks. The application was built and tested in the Development environment of Zoho Creator and replaces scattered spreadsheets and paper call sheets with a single, role-aware system for planning, tracking and handing over a film.

Keywords: Low-Code Development; Zoho Creator; Deluge Scripting; Film Production Management; Role-Based Access Control

PROJECT AT A GLANCE

Attribute	Details
Project Title	Film Production Management System
Application Domain	Media and entertainment – film production planning, scheduling and resource management
Student Name & Register No.	Menaha R – 240701308
Zoho Creator Edition	Zoho Creator free trial (premium features enabled)
Application Name (Link Name)	Film Production Management System (film-production-management-system)
Published / Live URL	https://creatorapp.zoho.in/menaha.r.2024.cse_rajalak147/film-production-management-system
Demo Credentials (per role)	Admin: menaha.r.2024.cse@rajalakshmi.edu.in Invited user: menaharajkumar2208@gmail.com
Number of Forms / Subforms	10 / 0
Number of Reports (by type)	List: 6 Spreadsheet: 1 Pivot: 1 Chart (bubble): 1
Number of Pages / Dashboards	1 (Home page dashboard)
Number of Workflows	Form: 10 Schedule: 3 Approval: 1 Blueprint: 1 Batch: 0
Deluge Functions / Custom Actions	5 functions (Calculate_Remaining_Budget, Calculate_Shooting_Days, Check_Shooting_Schedule, Movie_Category, Validate_Movie_Rating)
Integrations	Zia Build Agent (AI-assisted building); Zoho Creator e-mail notifications; Zoho Creator Portal (Film_Portal_User)
Roles & Profiles Defined	Roles: CEO, Producer, Director, Production Manager, Crew Member. Profiles: Admin, Producer, Director, Cinematographer, Crew Member, Post-Production Manager. Portal profile: Film_Portal_User
Environments Used	Development (Stage and Production publishing planned)
Date of Production Deployment	Not yet published (application is in the Development environment)

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CO	Course Outcome
CRUD	Create, Read, Update, Delete
CSV	Comma-Separated Values
ER	Entity–Relationship
FPMS	Film Production Management System
FR	Functional Requirement
ID	Identifier
JSON	JavaScript Object Notation
KPI	Key Performance Indicator
LCDP	Low-Code Development Platform
NFR	Non-Functional Requirement
PDF	Portable Document Format
RBAC	Role-Based Access Control
REST	Representational State Transfer
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Making a film is a project in the strictest sense. Every film brings together a temporary team of producers, directors, actors, technicians and daily-wage crew, who work for a fixed period on a fixed budget and then disperse. During that period the production has to coordinate many resources at once: the cast and crew who must be present, the locations that must be booked, the cameras, lights, microphones and cranes that must be rented and kept in working order, and the scenes that must be shot in a sensible order. A delay in any one of them, such as a damaged camera or an unavailable actor, can leave the whole unit idle, and an idle unit is expensive because crew wages and equipment rentals are paid for every day.

Chennai, where Rajalakshmi Engineering College is located, is home to a large regional film industry. Production houses of every size commonly manage their films with a mix of spreadsheets, paper call sheets, phone calls and messaging groups. The production manager keeps the crew list, the director keeps the scene breakdown, the producer keeps the budget, and the information is brought together only through conversations and end-of-day messages. As the number of films, people and shooting days grows, this arrangement becomes hard to keep accurate.

This project addresses that problem. It replaces scattered sheets and messages with a Zoho Creator application in which films, people, locations, equipment and scenes are recorded once as master data and then reused in schedules, equipment allocation, attendance and daily reports, and in which every user sees only what their role allows.

1.2 Problem Statement

Film production teams need to know, at any moment, which films are active, who is working on them, what is scheduled, what equipment is available and how much money has been spent. Under the current practice this information is spread across files and chat groups and is updated by hand, so it is often incomplete, repeated or out of date. A system is needed that records each production fact once, links it to everything that depends on it, and shows the right view to each member of the production team.

- Film, cast, crew, equipment and location details are written separately in several sheets, which leads to repeated entry and mismatched records.
- Shooting schedules are circulated as documents or chat messages, so a change of date, time or location may not reach everyone, and clashes between scenes, locations and people are noticed late.
- There is no dependable record of equipment condition or allocation; a camera that needs repair can still be planned for a shoot, and the rental cost per day is not visible to the producer.
- Crew attendance and daily progress are noted on paper or verbally, with no structured way to see who was late, absent or on half day, or why a scene was not completed.
- The producer cannot see budget, spending and progress across films in one place, and sensitive details such as remuneration are open to everyone who holds the file.

1.3 Objectives

- To design a normalised data model and forms for films, cast and crew, locations, equipment, scenes, schedules, equipment allocation, attendance and daily shooting reports in Zoho Creator.
- To connect the master forms to the transaction forms through lookup fields so that each fact is entered only once.
- To automate production control using a Blueprint for the film lifecycle, an approval process for equipment allocation and Deluge functions for budget, shooting-day and schedule calculations.
- To provide a home page dashboard and list, pivot and chart reports that give the producer and director a live view of films, equipment condition, rental cost and crew attendance.
- To implement role-based access with a role hierarchy and permission sets, and a portal permission through which actors and crew members can view their own details.
- To prepare the application for deployment through the Development, Stage and Production environments with governance controls.

1.4 Scope of the Project

In scope. The application is used by the production house team, with the roles of CEO, Producer, Director, Production Manager and Crew Member, and by actors and crew members through a restricted portal profile. It supports film registration with genre, language, director, producer, dates and status; maintenance of cast and crew with role, experience, daily remuneration and availability; location, equipment and scene masters; shooting schedules that link a film, a scene, a date, a time slot and a location; equipment allocation with approval; shooting attendance and its monitoring; daily shooting reports; and production completion and handover. A home page dashboard and a set of reports summarise the data. The application is used through the Zoho Creator web interface on a laptop and through the Zoho Creator mobile app on set.

Out of scope. The application does not handle script writing, editing or post-production file management, nor the payment of wages or rentals; remuneration and rental rates are recorded for planning and reporting only. Integration with an external accounting package and the automatic weekly e-mailing of a production budget report are planned as future enhancements. Casting auditions, ticketing and distribution of the finished film are also outside the project.

1.5 Justification for a Low-Code Approach

Film production management is a typical internal business process: a limited number of related entities, clear rules, approvals and role-based visibility. Zoho Creator provides the building blocks for exactly this kind of application — forms with lookups, reports, a page builder, the Deluge scripting language, form and scheduled workflows, approvals, Blueprints, user permissions and native web and mobile apps — on a hosted platform [1]. Hosting, authentication, the database and the mobile client did not have to be built, which let the effort go into the production logic itself: the data model, the schedule rules, the approval flow and the budget calculations.

Low-code platforms are intended to let developers and domain experts build applications through visual modelling with only limited hand-written code [2], [3]. That suited this project, which was built by one student within a single semester. Where the visual tools were not enough, Zoho Creator still allowed targeted code: Deluge functions for budget, schedule and rating logic [4], Blueprints and approvals for process control [5], and role and permission settings for access

control [6]. The application can therefore be maintained through configuration for routine changes while still supporting the custom behaviour a production house needs.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

The existing process for planning and running a film shoot in a typical production house works as follows:

1. The producer and director finalise the film plan: title, genre, language, budget and expected dates. These are kept in a document or a spreadsheet, and a new sheet is started for every film.

2. The production manager prepares the list of cast and crew, with phone numbers and daily rates, and confirms availability by calling people or posting in a messaging group.

3. The director's team breaks the script into scenes and prepares a shooting schedule, usually as a spreadsheet or a printed call sheet that names the date, time, scene and location.

4. The call sheet is forwarded by phone and chat. Equipment is booked from a rental house by phone, and its condition is known only from memory or from the rental house.

5. On the shooting day, attendance is noted on paper or by roll call, and an assistant writes a daily report describing the scenes completed, delays and problems, which reaches the producer at night.

6. Budget, spending and the handover of footage and equipment are reconciled informally by the accounts team and the production manager at the end of the shoot.

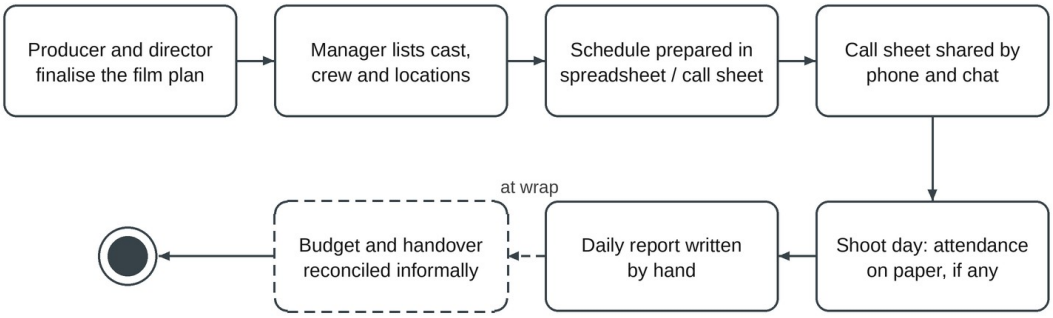


Figure 2.1: As-Is Process Flow of the Manual Film Production Process

2.2 Limitations of the Existing System

- Repeated data entry: the same cast, crew and film details are copied into several sheets for every film and every schedule.
- No single view: the producer cannot see films, schedules, equipment and budget together, so decisions depend on phone calls and memory.
- Weak schedule control: clashes of people, scenes and locations are found only on the day of the shoot, and changes depend on informal messages.
- No equipment history: damaged or repair-pending items are not tracked, and the rental cost per day is not summarised.
- Unstructured attendance and reports: late arrivals, half days and absences, and the reasons for incomplete scenes, are not recorded in a searchable form.
- Weak access control: files containing remuneration and budget are shared widely, with no way to limit what each role can view or change.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Several months (estimate) for a database, back-end API, web front-end, mobile app, authentication and hosting	About two weeks of part-time work in October 2026 for the complete application (actual for this project)
Skills required	Full-stack development: a back-end language and framework, database design, web and mobile front-end, DevOps	Zoho Creator configuration, Deluge scripting and an understanding of workflows and permissions
Hosting & infrastructure	Servers or cloud services must be provisioned, secured and maintained by the developer	Hosted by Zoho; separate Development, Stage and Production environments are available
Mobile application	A separate Android/iOS app would have to be built and published	The Zoho Creator mobile app runs the same application on the shooting location
Security & access control	Authentication, roles and data protection must be implemented by hand	Built-in user management, permission sets, role hierarchy, record-level criteria and portal permissions
Maintenance & change	Changes need code edits, testing and redeployment	Most changes are made in the builder and published through environments
Customisation limits	Unlimited; any UI or logic can be built	Limited to platform components; custom UI needs widgets, and some report options are restricted
Cost	Hosting and developer time; no platform licence	Per-user subscription after the free trial; no separate hosting cost

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	A. Sahay, A. Indamutsa, D. Di Ruscio and A. Pierantonio, 2020 [7]	Supporting the understanding and comparison of low-code development platforms (SEAA 2020)	Proposes a technical survey and a feature-based comparison of popular low-code development platforms, covering data modelling, UI design, business logic, integration and deployment.	Provided the vocabulary used to map this project's features (data model, workflows, roles, deployment) and to position Zoho Creator among LCDPs.
2	A. C. Bock and U. Frank, 2021 [2]	Low-Code Platform (Business & Information Systems Engineering, 63(6))	Characterises low-code platforms and discusses citizen developers, organisational agility and software development productivity.	Supports the justification that a single student developer can deliver a business-process application through a low-code platform.
3	D. Di Ruscio, D. Kolovos, J. de Lara, A. Pierantonio, M. Tisi and M. Wimmer, 2022 [3]	Low-code development and model-driven engineering: Two sides of the same coin? (Software and Systems Modeling, 21)	Compares low-code development with model-driven engineering, identifying their commonalities, differences, strengths and weaknesses.	Frames the forms, lookups and Blueprint used here as models from which Zoho Creator generates the running application.
4	M. A. A. Alamin, S. Malakar, G. Uddin, S. Afroz, T. B. Haider and A. Iqbal, 2021 [8]	An Empirical Study of Developer Discussions on Low-Code Software Development Challenges (arXiv:2103.11429)	Studies developer questions about low-code platforms on Stack Overflow; customisation is among the most discussed topics, and gaps in documentation are a recurring problem.	Matches the challenges expected in this project, such as going beyond the default layout and report components.

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
5	B. A. Bechky, 2006 [9]	Gaffers, Gofers, and Grips: Role-Based Coordination in Temporary Organizations (Organization Science, 17(1))	A field study of film sets showing that temporary crews coordinate their work through the expectations attached to roles rather than through personal familiarity.	Supports the role hierarchy and role-specific permission sets (CEO, producer, director, production manager, crew member) used in the application.
6	H. Blair, S. Grey and K. Randle, 2001 [10]	Working in film: Employment in a project based industry (Personnel Review, 30(2))	Describes film as a project-based industry in which most workers are hired for the length of a production and move from project to project.	Explains why availability, joining date and daily remuneration are recorded for every person in the Cast and Crew Master.
7	Zoho Corporation [1], [4], [5], [6]	Zoho Creator Help and Deluge Reference documentation	Official documentation of forms, reports, pages, workflows, approvals, Blueprints, permissions, environments and Deluge functions.	Primary technical reference for every component implemented in Chapter 5.

2.5 Emerging Trends in Low-Code Development

Three trends in low-code development shaped this project. The first is AI-assisted application building. Zoho Creator includes Zia, whose Build Agent can draft forms and workflows from a natural-language prompt. In this project the Build Agent was available as an assistant while the application was being built. Its output has to be reviewed, because the developer remains responsible for the final logic.

The second trend is citizen development combined with pro-code extension. Platforms increasingly let non-specialists build most of an application visually while professional developers extend it where necessary [2], [3]. The Film Production Management System follows that pattern: the forms, reports and home page are configured visually, while the budget, shooting-day, schedule, category and rating checks are small Deluge functions.

The third trend is role-based self-service and embedded analytics. Rather than sending reports by e-mail, low-code applications give each user a portal or dashboard that shows only the relevant data. Here the producer and director see summary cards, charts and pivot tables on the home page, while actors and crew members use a portal permission to view film details, schedules and assigned tasks.

2.6 Proposed System

The Film Production Management System is a Zoho Creator application that manages a film from registration to handover. It addresses each limitation of the existing system as follows. Repeated data entry is removed by master forms for films, cast and crew, locations, equipment and scenes, which are connected to schedules, allocations and reports through lookup fields. A single view is provided by a home page dashboard that shows the number of requests, the total budget, the number of members and the active projects, along with an equipment chart, a director search and a cast and crew board. Schedule control comes from the Shooting Schedule form, which ties a film, a scene, a date, a time slot and a location together, supported by the Check_Shooting_Schedule and Calculate_Shooting_Days functions. Equipment condition and rental cost per day are summarised in a pivot report, and every allocation request goes through an approval process. Attendance is recorded as present, late, half day or absent with remarks and is monitored in a bubble chart. The remaining budget is calculated by the Calculate_Remaining_Budget function. Finally, a role hierarchy, permission sets and a portal permission make sure that each person sees only the information needed for their work.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

The Film Production Management System is used by the people who plan and run a film. Their responsibilities were identified from the existing process described in Chapter 2 and are listed in Table 3.1. The roles match the role hierarchy and permission sets configured in the application (Section 5.8).

Table 3.1: Stakeholders and User Roles

Role	Stakeholder	Responsibilities in the Application
CEO / Producer	Production house owner or film producer	Registers films, sets budgets, reviews the dashboard and reports, approves equipment allocation
Director	Director and the directing team	Views project details, plans scenes and shooting schedules, updates production status
Production Manager	Person who runs the day-to-day unit	Maintains cast, crew, location and equipment masters, records attendance, manages post-production, completion and handover
Cinematographer	Camera and lighting head	Views shooting schedules, equipment allocation and film master records
Crew Member	Technicians and daily-wage crew	Views assigned tasks and shooting schedules, updates task status
Portal User	Actors and crew members	Views their own film details, schedules and assigned tasks through the portal permission

3.2 Functional Requirements

The functional requirements, derived from the objectives in Section 1.3 and the limitations in Section 2.2, are given in Table 3.2. Each requirement is traced to the section of Chapter 5 in which it is implemented.

Table 3.2: Functional Requirements

ID	Requirement	Implemented in
FR1	The system shall record a film with its name, genre, language, director, producer, start date and description.	Section 5.2

ID	Requirement	Implemented in
FR2	The system shall maintain a master of cast and crew with role, contact details, experience, daily remuneration, availability status and joining date.	Section 5.2
FR3	The system shall maintain masters for locations, equipment and scenes.	Section 5.2
FR4	The system shall create a shooting schedule that links a film, a scene, a date, a start and end time and a location through lookup fields.	Section 5.2
FR5	The system shall record equipment allocation and route each allocation request through an approval.	Sections 5.2, 5.6
FR6	The system shall record shooting attendance as present, late, half day or absent with remarks.	Section 5.2
FR7	The system shall capture a daily shooting report and the production completion and handover details.	Section 5.2
FR8	The system shall move a film through its life cycle using a Blueprint.	Section 5.6
FR9	The system shall calculate remaining budget, shooting days, schedule status, movie category and rating checks using Deluge functions.	Section 5.5
FR10	The system shall provide list, pivot, chart and spreadsheet reports for films, cast and crew, equipment, schedules and attendance.	Section 5.3
FR11	The system shall provide a home page dashboard showing requests, total budget, members, active projects and an equipment chart.	Section 5.4
FR12	The system shall restrict each user to the forms and reports allowed by the role and permission set.	Section 5.8
FR13	The system shall let actors and crew members view their own details through a portal permission.	Sections 5.8, 5.11
FR14	The system shall be usable from a web browser and the Zoho Creator mobile app.	Section 5.11

3.3 Non-Functional Requirements

Non-functional requirements describe the qualities expected of the application rather than its features. They are summarised in Table 3.3.

Table 3.3: Non-Functional Requirements

ID	Attribute	Requirement
NFR1	Usability	A user with no technical background shall be able to add a record from a form and find it in a report without training; the sidebar groups every form and report under its master.
NFR2	Security	Each role shall see only the forms and reports granted by its permission set; remuneration and budget data shall not be open to every user.
NFR3	Data integrity	Related data shall be entered once and reused through lookup fields; mandatory fields shall be enforced on forms.
NFR4	Performance	Lists, pivots and the dashboard shall load without a noticeable delay for the data volume of a single production house.
NFR5	Availability	The application shall be hosted on Zoho Creator and reachable from any location with an internet connection.
NFR6	Portability	The same application shall run on a laptop browser and on the Zoho Creator mobile app without separate development.
NFR7	Maintainability	Routine changes shall be made through configuration in the builder; custom logic shall be kept in small, named Deluge functions.
NFR8	Scalability	New films, people and equipment shall be added as records without changing the design.

3.4 Hardware and Software Requirements

Because the application is hosted on Zoho Creator, the requirements on the user side are modest. Table 3.4 lists what was used for development and what an end user needs.

Table 3.4: Hardware and Software Requirements

Item	Specification
Development device	Laptop or desktop with a modern web browser (Google Chrome) and a stable internet connection
Development platform	Zoho Creator 6 (India data centre), trial subscription; Development environment
Scripting language	Deluge, the scripting language built into Zoho Creator
AI assistance	Zia Build Agent inside the Creator builder
End-user device (web)	Any laptop or desktop with a current browser
End-user device (mobile)	Android or iOS smartphone with the Zoho Creator mobile app
Report preparation	Microsoft Word and Python (matplotlib) for the report and diagrams
Network	Broadband or mobile data; no server or database software needs to be installed

3.5 Use Case Model

Figure 3.1 shows the main actors and the use cases they perform. The Producer and the Production Manager are the heaviest users of the system, the Director concentrates on schedules, and crew members and portal users mostly read information that concerns them and record attendance and reports.

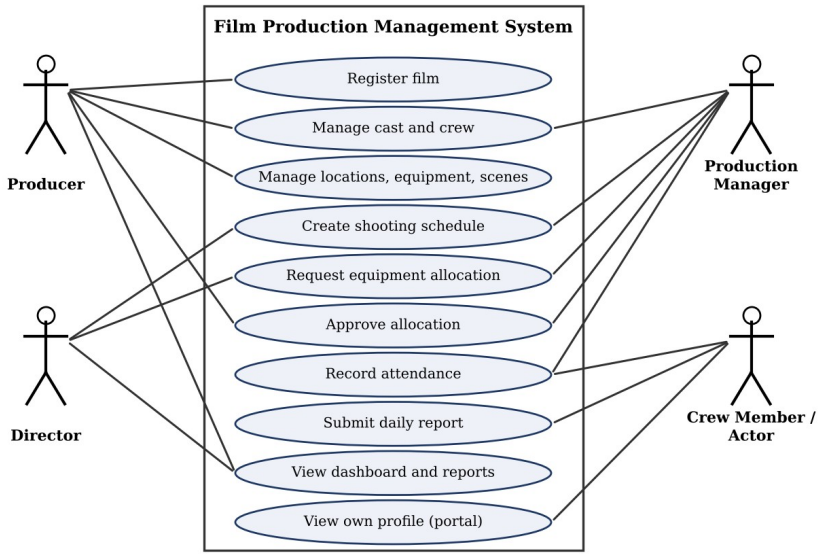


Figure 3.1: Use Case Diagram of the Film Production Management System

The most important use case, scheduling a shoot and recording attendance, is described in Table 3.5.

Table 3.5: Use Case Description – Schedule a Shoot and Record Attendance

Item	Description
Use case	Schedule a shoot and record attendance
Actors	Director, Production Manager, Crew Member
Precondition	The film, scene, location and cast and crew records already exist in the master forms; the user is logged in with a role that permits the action.
Main flow	1. The Director opens the Shooting schedule form. 2. The film, scene and location are chosen from lookup lists. 3. The shooting date, start time and end time are entered and the form is submitted. 4. The schedule appears in All Schedules. 5. On the shooting day the Production Manager opens the Shooting Attendance form and records the status of each crew member. 6. The attendance appears in All Attendances and in the attendance monitoring chart.
Alternative flow	If a required field such as the schedule ID or the start time is empty, the form is not submitted and the field is highlighted.
Postcondition	The schedule and attendance records are stored and visible in the reports to the permitted roles.

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

The application follows the layered structure that Zoho Creator provides. Users reach it through a browser or the mobile app. The presentation layer consists of the sidebar navigation, forms, reports and the home page. The logic layer holds the form workflows, the Blueprint, the approval and the Deluge functions. The data layer is the Creator database, in which each form is a table. The security layer applies the role hierarchy, permission sets and portal permission to every request. Hosting, backup and the Development, Stage and Production environments are supplied by the platform. Figure 4.1 shows the arrangement.

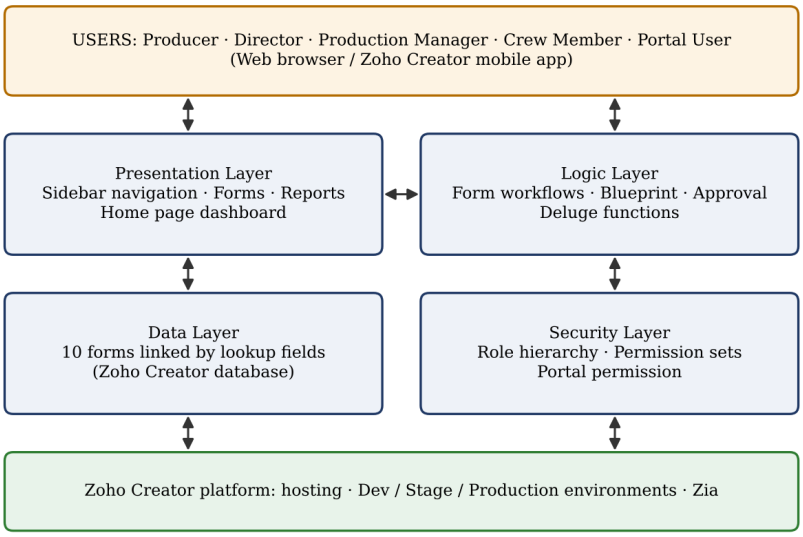
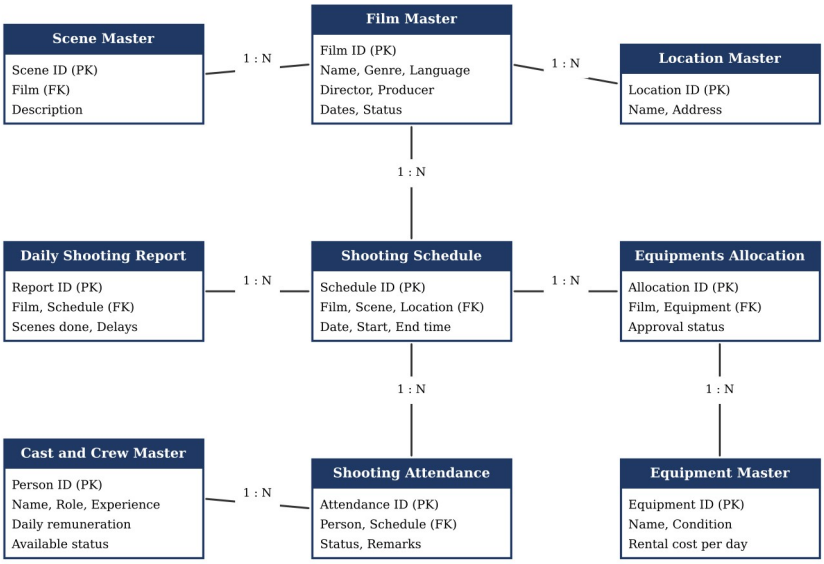


Figure 4.1: System Architecture of the Film Production Management System

4.2 Data Model

The data model has ten forms: nine of them are shown in Figure 4.2, and Production Completion and Handover is attached to the film. Films, people, locations, equipment and scenes are master forms. Shooting Schedule, Equipments Allocation, Shooting Attendance and Daily Shooting Report are transaction forms that refer to the masters through lookup fields, so a fact is stored once and displayed wherever it is needed. No subforms are used.



Production Completion and Handover links to Film Master through a lookup field (not shown).

Figure 4.2: Entity–Relationship Diagram of the Film Production Management System

The relationships shown in the diagram are listed in Table 4.1. The relationships are inferred from the lookup fields on the forms.

Table 4.1: Entities and Relationships

Parent Entity	Child Entity	Cardinality	Linked by
Film Master	Scene Master	1 : N	Lookup field
Film Master	Shooting Schedule	1 : N	Lookup field
Scene Master	Shooting Schedule	1 : N	Lookup field
Location Master	Shooting Schedule	1 : N	Lookup field
Shooting Schedule	Shooting Attendance	1 : N	Lookup field
Cast and Crew Master	Shooting Attendance	1 : N	Lookup field
Equipment Master	Equipments Allocation	1 : N	Lookup field
Film Master	Equipments Allocation	1 : N	Lookup field
Shooting Schedule	Daily Shooting Report	1 : N	Lookup field

4.3 Form Design

Each form uses the standard Creator field types so that validation and reports work without extra code. The auto-number ID fields identify records, dropdowns restrict values such as role and condition, and lookups restrict references to existing masters. The field specifications of the five main forms are given in Tables 4.2 to 4.6. Fields shown in the screenshots are listed exactly; the remaining details are marked for checking.

Table 4.2: Field Specification – Film Master Form

Field	Type	Notes
Film ID	Number	Unique identifier, for example 101 to 110
Film Name	Single line	Title of the film
Genre	Dropdown	Action, Comedy, Drama, Horror, Romance, Thriller
Language	Dropdown	Tamil, Telugu, Malayalam, Hindi, English
Director's name	Single line	Name of the director
Producer name	Single line	Name of the producer
Start date	Date	Format dd-MMM-yyyy
Description	Multi line	Short story line

Table 4.3: Field Specification – Cast and Crew Master Form

Field	Type	Notes
Person ID	Number	Identifier, seven digits (#######)
Full name	Single line	Mandatory
Role	Dropdown	Actor, director, camera and other roles
Mobile number	Phone	Country code +91 by default
Email	Email	Contact address
Experience	Number	Years of experience
Daily Remuneration	Currency	Rate per day, format ##,##,###.##
Available Status	Dropdown	Whether the person is free for a shoot
Joining date	Date	Format dd-MMM-yyyy
Remarks	Multi line	Free notes

Table 4.4: Field Specification – Shooting Schedule Form

Field	Type	Notes
schedule ID	Number	Mandatory identifier
Film Master	Lookup	Chooses a film from Film Master
Scene Master	Lookup	Chooses a scene from Scene Master
shooting date	Date	Format dd-MMM-yyyy
start time	Time	Mandatory; format HH:mm:ss
End time	Time	Format HH:mm:ss
Location Master	Lookup	Chooses a location from Location Master

Table 4.5: Field Specification – Equipment Master Form

Field	Type	Notes
Equipment name	Single line	For example Boom Microphone, Jib Crane, Tripod
Condition	Dropdown	good, need repair or damaged
Rental cost per day	Currency	Summed in the pivot report

Table 4.6: Field Specification – Shooting Attendance Form

Field	Type	Notes
Crew member	Lookup	Chooses a person from Cast and Crew Master
Attendance Status	Dropdown	Absent, Half Day, Late or Present
Remarks	Single line	Reason, for example late pack-up or scene completed
Shooting date / schedule	Lookup	Links the record to a schedule

The remaining forms follow the same pattern. Location Master stores the place and its details, Scene Master stores the scene and its film, Equipments Allocation links a film to an item of equipment and carries the approval status, Daily Shooting Report records the progress of the day, and Production Completion and Handover records the closing of the film.

4.4 Process and Workflow Design

A film passes through the stages shown in Figure 4.3. The producer registers the film, the director links scenes and locations, a schedule is created and equipment is requested. The request goes through the equipment approval. On the shooting day attendance is recorded and the daily report is submitted, and at the end the production is marked complete and handed over. The home page and the reports are updated automatically at every step because they read the same records.

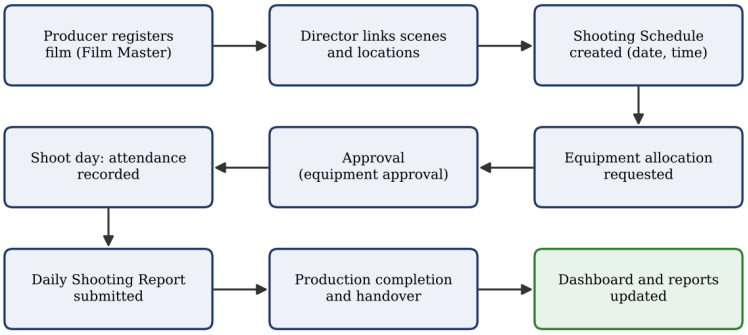


Figure 4.3: Process Flow for Shooting Schedule and Daily Report

Two mechanisms control the process. The Movie Production Blueprint governs the stages of the film and the shooting schedule, and the equipment approval governs the use of equipment. Both are described in Section 5.6.

4.5 User Interface and Navigation Design

The sidebar groups the application by master: Film Masters, Approvals, Cast and Crew Master, Location Masters, Equipment Masters, Scene Masters, Home page, Schedules, Equipments allocation, Daily Shooting Reports and Production Completion and Handover, with Attendances at the end. Each group opens to its form and its reports, so the user reaches an entry form and its list in one click. The home page is a dashboard of summary cards that gives the producer an overview on arrival. Figure 4.4 shows which parts of the navigation each role uses.

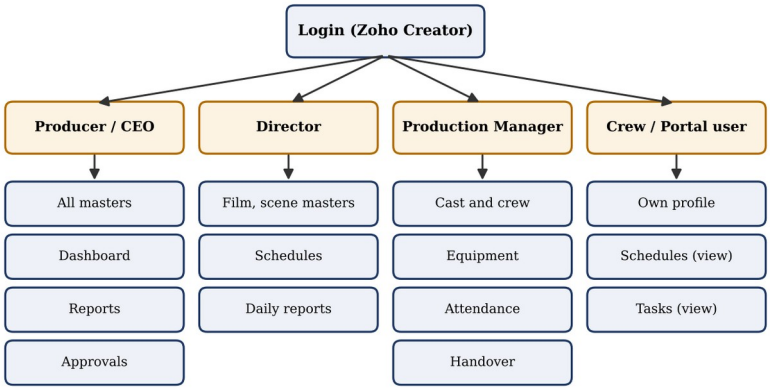


Figure 4.4: Navigation Structure of the Application by Role

4.6 Access Control Design

Access is designed in three layers. First, the role hierarchy places the CEO at the top, followed by the producer, then the director and the production manager, and the crew member at the bottom, so that a person higher in the hierarchy can see the records of those below. Second, six permission sets (Admin, Producer, Director, Cinematographer, Crew Member and Post-Production Manager) define what each type of user may do. Third, a portal permission, Film_Portal_User, gives actors and crew members a restricted view of their own details. Table 4.7 summarises the intended permissions, which are derived from the descriptions of the permission sets.

Table 4.7: Role–Permission Matrix

Area	Admin	Producer	Director	Crew Member	Post-Production Mgr
Forms, reports and workflows (full)	Yes	No	No	No	No
Film projects and budgets	Yes	Manage	View	No	View
Cast and crew, locations	Yes	Manage	View	No	No
Shooting schedules	Yes	Manage	Manage	View	View
Equipment allocation	Yes	Manage	View	No	View
Task and attendance status	Yes	View	Update	Update	No
Post-production, completion, daily reports	Yes	View	View	No	Manage

The Cinematographer permission set additionally allows camera operations, lighting and visual composition, with access to shooting schedules, equipment allocation and film master records.

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application, named Film Production Management System, was created in Zoho Creator 6 under a trial subscription (India data centre) and is currently in the Development environment. It was built using the Design, Workflow and Settings areas of the builder, with the Zia Build Agent available as an assistant. Figure 5.1 shows the application as the user sees it, with the Film Masters group expanded to its form and two reports.

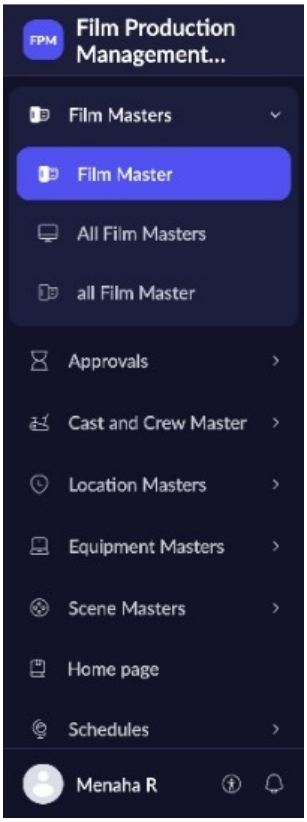


Figure 5.1: Application Home – Navigation Sidebar with the Film Masters Section Expanded

In total the application contains ten forms, nine reports and one page. Their names are listed in Table 5.1 (reports) and in the sidebar of Figure 5.1.

5.2 Forms and Data Capture

Forms were created in the form builder by dragging fields from the Basic Fields palette and setting their properties. For every field the display name, the field link

name, the display format, validation (mandatory and no duplicate values), initial value and maximum digits can be set. For example, the Person ID field of Cast and Crew Master is a number with ten digits, and Full name is mandatory. Figure 5.2 shows the form in the builder and in live mode.

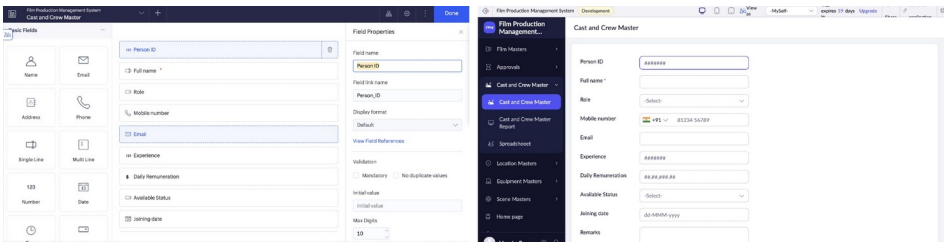


Figure 5.2: Cast and Crew Master Form – (left) in the Form Builder, (right) in Live Mode

The Shooting schedule form, shown in Figure 5.3, uses three lookup fields. The Film Master, Scene Master and Location Master dropdowns are filled from the master forms, so the user selects an existing film, scene and location instead of typing them again. The schedule ID and the start time are mandatory. The submit button is labelled Shooting.

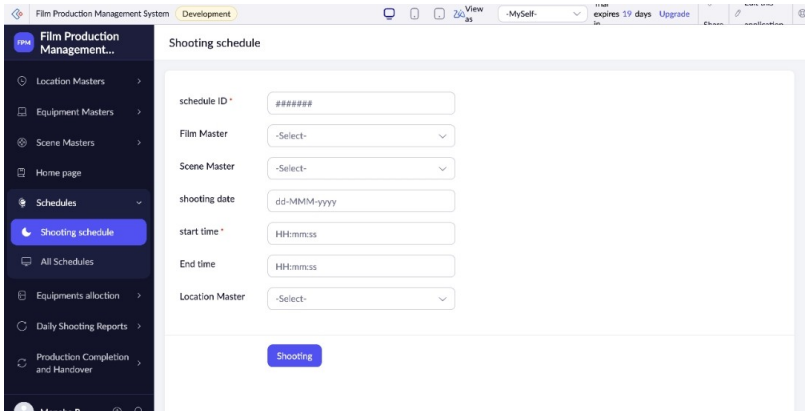


Figure 5.3: Shooting Schedule Form in Live Mode

The other forms (Film Master, Location Master, Equipment master, Scene Master, Equipments allocation, Daily Shooting Report, Shooting Attendance Form and Production Completion and Handover) were built in the same way. Sample data of ten films, ten equipment items and a set of crew members was entered to test the reports and the dashboard.

5.3 Reports

Reports present the stored records in the form most useful to each reader. Nine reports were built, as listed in Table 5.1.

Table 5.1: Reports Implemented

No.	Report	Type	Purpose
1	All Film Masters	List	All films with genre, language, director, producer and start date
2	all Film Master	List	Second list report on the Film Master form
3	Cast and Crew Master Report	List	People with role, experience, remuneration and availability
4	Spreadsheet (Cast and Crew)	Spreadsheet	Spreadsheet-style view of the cast and crew data
5	All Equipment Masters	List	All equipment with condition and rental cost
6	pivot table	Pivot	Rental cost per day by equipment name and condition
7	All Schedules	List	All shooting schedules
8	All Attendances	List	All attendance records
9	attendance monitoring	Bubble chart	Attendance status of each crew member

Figure 5.4 shows the film list and the equipment pivot. The film list reports ten records (“Showing 10 of 10”) and allows search, adding a record and further actions from its toolbar. The pivot report groups equipment by condition and totals the daily rental: Rs.7,500 for damaged items, Rs.202,500 for good items and Rs.26,000 for items needing repair, a grand total of Rs.236,000 per day. This lets the producer see at once how much of the rental cost is tied to equipment that cannot be used.

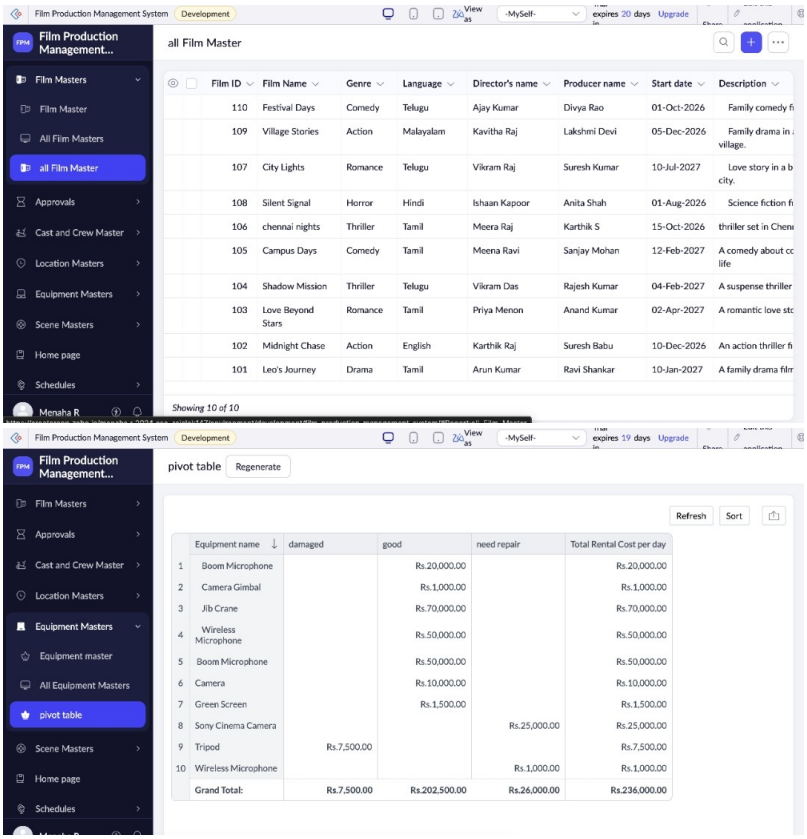


Figure 5.4: All Film Masters List Report and Equipment Pivot Report

5.4 Pages and Dashboards

The Home page was built with the page builder as a dashboard. It has summary cards for All requests (1,055), Total Budget, Total members and Active projects (10), a doughnut chart of equipment, a search box to find a director by name and a cast and crew board. Figure 5.5 shows the page.

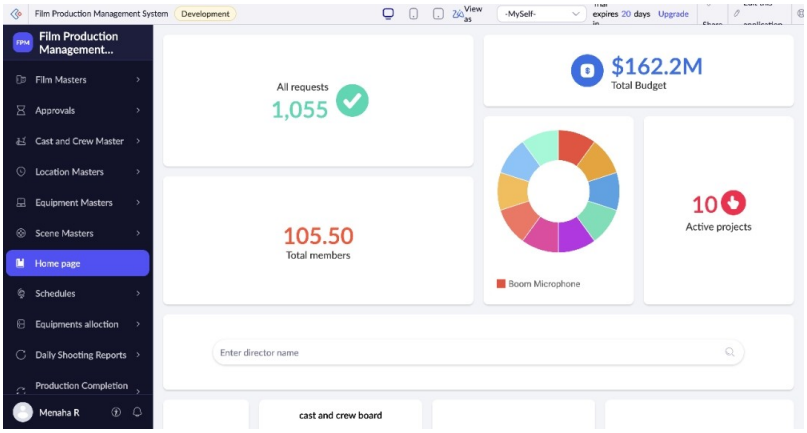


Figure 5.5: Home Page Dashboard

5.5 Deluge Scripting

Deluge is the scripting language of Zoho Creator. Five custom functions were written in the Workflow area under the Functions tab, as listed in Table 5.2.

Table 5.2: Deluge Scripts Implemented

Function	Language	Returns	Purpose
Calculate_Remaining_Budget	Deluge	float	Returns the budget left after spending
Calculate_Shooting_Days	Deluge	int	Counts the number of shooting days
Check_Shooting_Schedule	Deluge	string	Checks a schedule and returns a message
Movie_Category	Deluge	string	Returns the category of a movie
Validate_Movie_Rating	Deluge	string	Validates the rating entered for a movie

The functions are defined in the Default namespace. Only Calculate_Remaining_Budget was opened in the editor, and its source is given in Code 5.1.

Code 5.1: Calculate_Remaining_Budget Function

```
float Calculate_Remaining_Budget()
{
    total = 5000000;
    spent = 2000000;
    remaining = total - spent;
    return remaining;
}
```

The function subtracts the amount spent from the total budget and returns the remaining amount of 3,000,000. In this version the two amounts are fixed in the code. A production version would read them from the Film Master and Daily Shooting Report records using fetch and aggregate statements, which is listed as a future enhancement. The editor, with its palette of Deluge statements (conditions, data access and others), and the list of functions are shown in Figure 5.6.

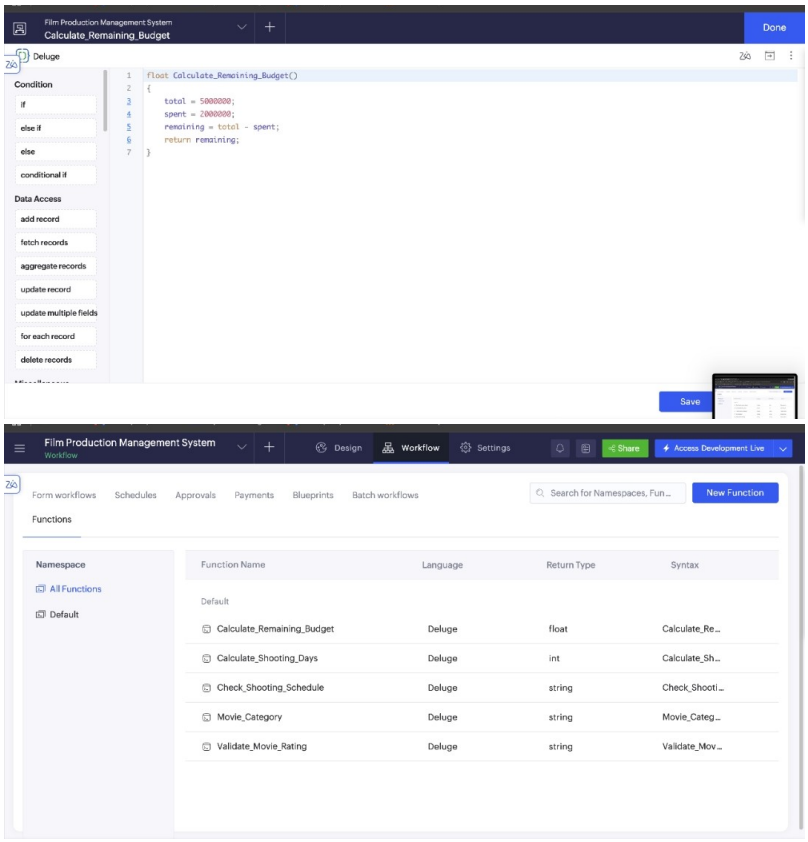


Figure 5.6: Calculate_Remaining_Budget Function and the List of Deluge Functions

5.6 Workflows and Automation

Automation was added through a Blueprint and an approval. The Blueprint named Movie Production is enabled for the Film Master and Shooting schedule forms and was created on 30 October 2026. It controls the stages through which a film or a schedule moves and permits only the allowed transitions. The approval named equipment approval is attached to the Equipments allocation form, is triggered on review and has four actions. Both are shown in Figure 5.7 and summarised in Table 5.3.

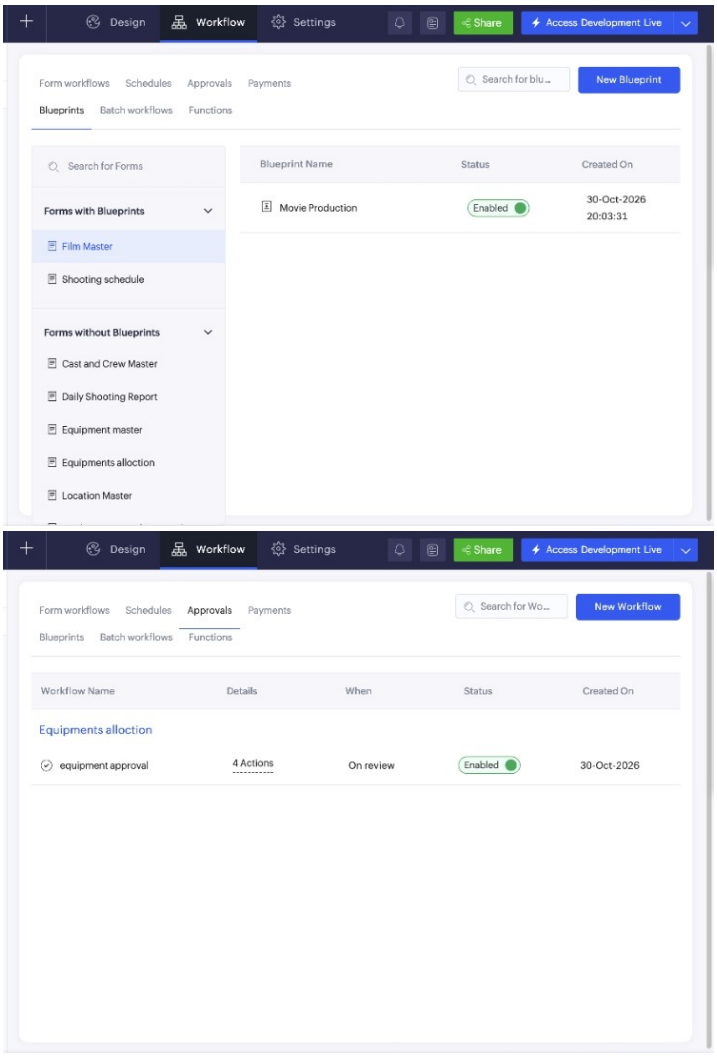


Figure 5.7: Movie Production Blueprint and Equipment Approval Workflow

Table 5.3: Workflows Implemented

Workflow	Applies to	Status	Behaviour
Movie Production (Blueprint)	Film Master, Shooting schedule	Enabled	Moves a record through defined stages
equipment approval (Approval)	Equipments allocation	Enabled	On review; four actions after approval or rejection
Form workflows	—	Not recorded	No form workflow was captured in the screenshots of this version

5.7 Integrations

The application is self-contained and relies on the services supplied inside Zoho Creator. No third-party REST API was connected. The services used are listed in Table 5.4.

Table 5.4: Integrations and Platform Services Used

Service	Use in the Application
Zoho Creator workflows	Blueprint and approval for production control
Zia Build Agent	AI assistant available in the builder while the application was being built (Section 2.5)
Zoho Creator mobile app	Access to the application on the shooting location
Portal	Access for actors and crew members
Environments	Development in use; Stage and Production prepared for publishing
External integrations	None configured

5.8 User Management and Role-Based Access

Users are managed in the Zoho Creator admin console. Two users were added: Menaha R, the super admin, and a second user invited by e-mail. The role hierarchy and the permission sets are defined under Settings, User Permissions. The hierarchy is CEO, then producer, then director and production manager, then crew member. Six permission sets were created: Admin, Cinematographer, Crew Member, Director, Post-Production Manager and Producer. Figure 5.8 shows the hierarchy and the permission sets.

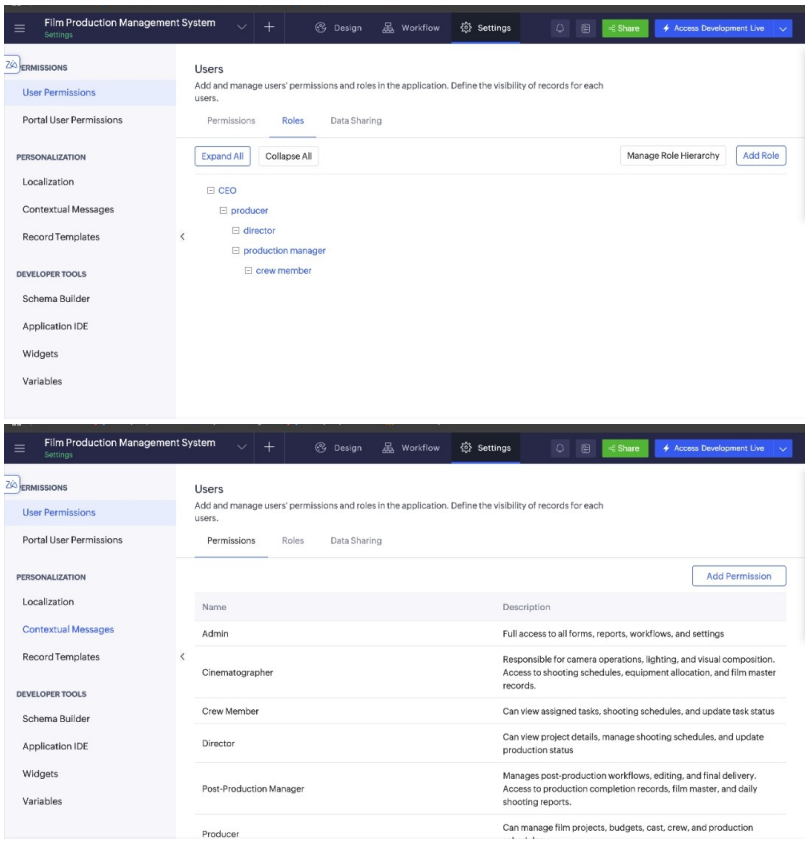


Figure 5.8: Role Hierarchy and User Permission Sets

5.9 AI and Analytics Features

Two features serve this section. The first is the Zia Build Agent, the AI assistant of the builder described in Section 2.5. The second is analytics: the attendance monitoring bubble chart plots each crew member against the attendance status (Absent, Half Day, Late or Present), with each bubble labelled by the remark entered. The chart can be refreshed, sorted and exported, and it was generated with the Regenerate option. It shows at a glance which crew members were absent, late, on half day or present, with remarks such as ‘scene completed’ and ‘late pack-up’. Figure 5.9 shows the chart together with the dashboard of Section 5.4 and the pivot of Section 5.3, which form the analytics layer of the application.

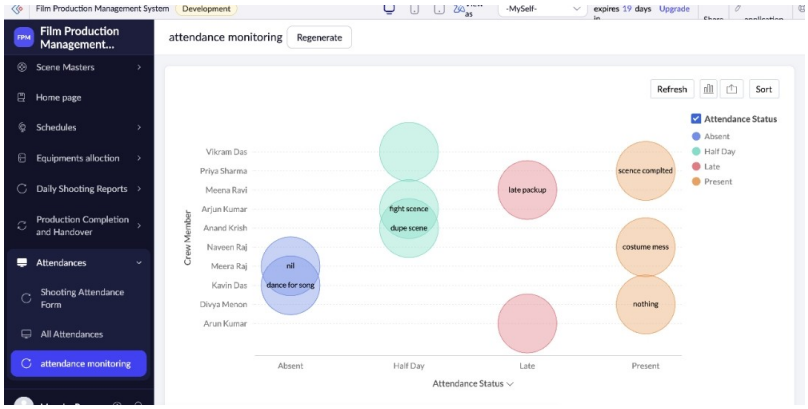


Figure 5.9: Attendance Monitoring Bubble Chart

5.10 Pro-Code Extension

The pro-code part of the project is the Deluge code of Section 5.5. Zoho Creator also offers the Application IDE, widgets and variables for deeper extension, and these were available under Developer Tools in Settings. The project did not require them because the default components covered the requirements, which is consistent with the low-code approach justified in Section 1.5.

5.11 Mobile and Portal Access

The application runs on the Zoho Creator mobile app without any separate development; the forms and reports adapt to the smaller screen, which lets crew record attendance on location. For actors and crew members who do not belong to the production team, a portal was set up. The portal permission Film_Portal_User gives access to film details, shooting schedules and assigned tasks, and the default Customer profile allows only adding and viewing own data. Figure 5.10 shows the portal permissions and the users list, in which the owner is the super admin and the second user is still in the invited state.

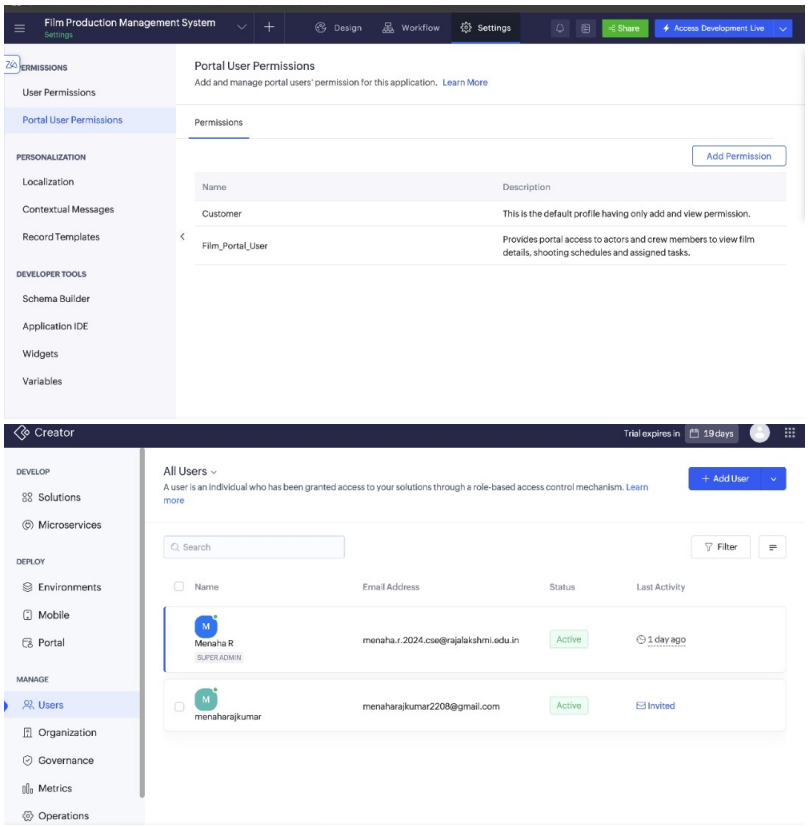


Figure 5.10: Portal User Permissions and the Users List

This completes the implementation. Testing and debugging of the application is presented in Chapter 6.

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was carried out at four levels while the application was being built in the Development environment. Unit testing checked each form and each Deluge function in isolation: every form was opened in live mode, a record was added and its fields, dropdowns, lookups and mandatory markers were checked, and each custom function was run with the Execute option of the function editor. Integration testing covered the paths that cross components: a master record appearing in the lookup of the Shooting schedule form, an equipment allocation entering the equipment approval, a film or schedule moving through the Movie Production Blueprint, and the home page cards, pivot and charts reading the records entered in the forms.

Role-based access testing used the View as option of the live-mode toolbar together with the role hierarchy and the six permission sets described in Section 5.8, to check what each type of user can open. User acceptance testing is recorded in Section 6.5. Sample data of ten films (Film ID 101 to 110), ten equipment items and a set of crew members was used throughout, and every test case is traced to a functional requirement of Table 3.2 in Table 6.1.

6.2 Test Cases

Table 6.1 lists the twenty test cases that were designed for the application. Cases whose result can be seen in the screenshots of Chapters 5 and 8 are marked Pass or Fail with the evidence quoted. Cases marked To verify are role or device tests that depend on the second user accepting the invitation or on a mobile screenshot, and are to be completed before submission.

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR1	Register a film	Open Film Master, fill Film ID, name, genre, language, director and producer; submit	Record saved and listed in All Film Masters	Ten films (101–110) listed, “Showing 10 of 10” (Figure 5.4)	Pass
TC-02	FR1	Mandatory fields on Film Master	Submit the form with Film Name empty	Submission blocked; field highlighted	Film ID, Film Name, Genre and Film status carry the mandatory marker (Figure 8.1)	Pass
TC-03	FR2	Add a cast or crew member	Open Cast and Crew Master; enter full name, role, mobile, remuneration and joining date	Record saved with +91 as default country code	Form shows Person ID, +91 default and mandatory Full name (Figure 5.2)	Pass
TC-04	FR3	Add equipment with condition and rental cost	Add ten equipment items with condition and rental cost per day	Items listed and totalled in the pivot report	Ten items grouped by good, need repair and damaged (Figure 5.4)	Pass
TC-05	FR4	Shooting schedule using lookups	Open Shooting schedule; open the Film Master, Scene Master and Location Master dropdowns	Existing masters offered; no retyping	Three lookup dropdowns shown; button labelled Shooting (Figure 5.3)	Pass
TC-06	FR4	Mandatory schedule fields	Submit a schedule without schedule ID or start time	Submission blocked	schedule ID and start time marked mandatory (Figure 5.3)	Pass
TC-07	FR5	Equipment allocation approval	Submit an Equipments allocation record	Record routed to the equipment approval (On review)	Approval enabled with four actions, triggered on review (Figure 5.7)	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-08	FR6	Record shooting attendance	Open Shooting Attendance Form; choose crew member, status and remarks	Record listed in All Attendances and plotted in the chart	Absent, Half Day, Late and Present bubbles with remarks (Figure 5.9)	Pass
TC-09	FR7	Daily report and completion forms	Open Daily Shooting Reports and Production Completion and Handover	Forms and reports available from the sidebar	Both groups present in the sidebar (Figure 5.1)	Pass
TC-10	FR8	Movie Production Blueprint	Open a Film Master or Shooting schedule record and perform a transition	Record moves only to an allowed stage	Blueprint enabled for both forms (Figure 5.7)	To verify
TC-11	FR9	Remaining budget function	Execute Calculate_Remaining_Budget	Returns total minus spent	Returns 3,000,000 (5,000,000 − 2,000,000) (Code 5.1)	Pass
TC-12	FR9	Other four functions	Execute the four other functions of Table 5.2 with sample values	int and string results as listed in Table 5.2	Not run in this version; only Calculate_Remaining_Budget was executed (TC-11)	To verify
TC-13	FR10	Film list report	Open all Film Master; use search and sort	All records shown; toolbar actions work	Ten records with search, add and more actions (Figure 5.4)	Pass
TC-14	FR10	Equipment pivot totals	Open pivot table report	Totals equal the sum of the condition columns	Rs.7,500 + Rs.202,500 + Rs.26,000 = Rs.236,000 per day (Figure 5.4)	Pass
TC-15	FR10	Attendance monitoring chart	Open the chart; use Refresh and Sort	Chart redraws with current data	Bubble chart drawn for ten crew members (Figure 5.9)	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-16	FR11	Home page dashboard	Open Home page	Cards show correct counts and amounts	Cards shown, but Total members reads 105.50 and Total Budget uses $ (Figure 5.5); logged as BUG-02	Fail
TC-17	FR12	Role hierarchy and permission sets	Open Settings, User Permissions	Hierarchy and six permission sets present	CEO → producer → director / production manager → crew member; six sets (Figure 5.8)	Pass
TC-18	FR12	Crew member view	Use View as with a crew member account	Only assigned tasks and schedules visible	Not completed; the second user has not yet accepted the invitation (BUG-08)	To verify
TC-19	FR13	Portal access	Sign in as a portal user with Film_Portal_User	Own film details, schedules and tasks only	Portal permission defined (Figure 5.10); second user still invited	To verify
TC-20	FR14	Mobile app access	Open the application in the Zoho Creator mobile app	Forms and reports adapt to the screen	Not completed; no mobile screenshot was captured in this version	To verify

Of the twenty cases, fourteen passed, one failed and the remaining five depend on activities that were not completed at the time of writing. The failure is analysed in Section 6.4.

6.3 Debugging Deluge Scripts

Two techniques were used to find and correct problems in the Deluge functions. First, the Execute option of the function editor was used to run a function on its own and read the returned value, which for Calculate_Remaining_Budget confirmed the result of 3,000,000. Second, the line numbers and error messages of the editor were read after every Save; a statement that the editor could not validate is flagged against its line, which is how a mistyped variable or a missing semicolon is found.

Figure 6.1 shows the first technique applied to Validate_Movie_Rating. The function was executed with a test rating of 8; the editor reported Executed Successfully and returned the value Valid Movie Rating.

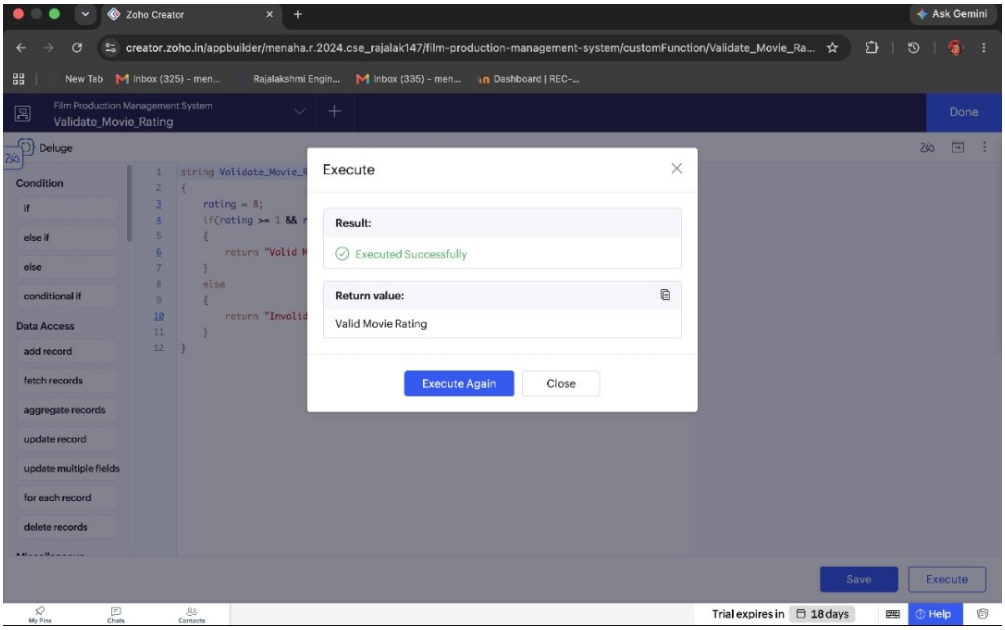


Figure 6.1: Executing the Validate_Movie_Rating Function in the Deluge Function Editor

A representative case is BUG-01. Executing Calculate_Remaining_Budget returned 3,000,000 every time, because the total (5,000,000) and the amount spent (2,000,000) are fixed values in the code (Code 5.1). The function runs without any error and still gives a wrong answer for a real film, which shows that a script that validates is not necessarily correct and that its result has to be compared with the data it is meant to read. The correction is to read the budget from Film Master and the expenses from Daily Shooting Report with fetch and aggregate statements.

6.4 Defect Log

Defects were recorded as they were found in the screens and in the test cases. Table 6.2 lists eight of them with the cause and the correction. They are still open at the time of writing; each will be corrected in the Development environment and re-tested before the application is published (Section 7.1).

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied / Planned	Status
BUG-01	Calculate_Remaining _Budget always returns 3,000,000	Total (5,000,000) and spent (2,000,000) are fixed values in the code	Read the budget from Film Master and the expenses from Daily Shooting Report using fetch and aggregate statements	Open
BUG-02	Home page shows Total members as 105.50 and Total Budget as $162.2M	Card bound to a decimal field summed instead of a record count; currency format not set to rupees	Change the card to the count of Cast and Crew Master records and set the currency format to Rs.	Open
BUG-03	Two similar list reports on Film Master: All Film Masters and all Film Master	A second report was created while building and left in the menu	Delete one report or give each a distinct purpose	Open
BUG-04	Boom Microphone and Wireless Microphone each appear twice in the equipment pivot	Equipment name is free text, so the same item can be entered twice with different rates	Add an equipment ID and quantity, and set No duplicate values on the name	Open
BUG-05	Spelling mistakes in labels and sample data, for example “Equipments allloction”, “Spreadsheeet”, “scence complted”	Names and remarks typed manually	Correct the form, report and menu names and the sample remarks	Open
BUG-06	Bubbles overlap in the attendance monitoring chart	Several crew members share the same status and remark	Replace with a stacked bar chart of count by status, or aggregate by crew member	Open
BUG-07	Film list is not in Film ID order (110, 109, 107, 108 …)	Report sorted by the time the record was added	Set the sort of the report to Film ID ascending	Open
BUG-08	Role tests for the second user cannot be completed	The user menaharajkumar is still in the Invited state	Accept the invitation and repeat TC-18 and TC-19	Open

6.5 User Acceptance Testing

No separate group of users was available within the trial period, so acceptance testing was carried out by the developer, who worked through the tasks of each role end to end on the web version in the Development environment. Table 6.3 records the problems noticed from the user's point of view and the action taken or planned for each; the related defects are listed in Table 6.2.

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
Menaha R (menaha.r.2024.cse@rajalakshmi.edu.in), super admin	Producer / Admin	Dashboard card values looked unusual; two similar film reports in the menu; equipment names repeated in the pivot	Logged as BUG-02, BUG-03 and BUG-04; corrections planned
menaharajkumar (menaharajkumar2208@gmail.com)	Crew Member or portal user	Invitation not yet accepted, so the restricted view could not be observed	Logged as BUG-08; to be repeated after acceptance

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

Zoho Creator provides Development, Stage and Production environments. Changes are made in Development, published to Stage for verification as a versioned release, and then published to Production, where end users work [9]. The Film Production Management System was built entirely in the Development environment, which is shown in the header of every live-mode screen in Chapter 5. The deployment procedure is: open Environments under Deploy, choose Publish to Stage, select the application and its changed components, enter a title and a description; after the Stage version has been verified, choose Publish to Production and select that version. Because a published version cannot be rolled back, the rollback plan is to publish a corrected version and to keep an exported copy of each released version as a backup.

Figure 7.1 shows the Environments page. The application appears in the list with the label Changes available, and the Stage and Production columns show the published versions 1.2 and 1. The application has therefore been published to both environments.

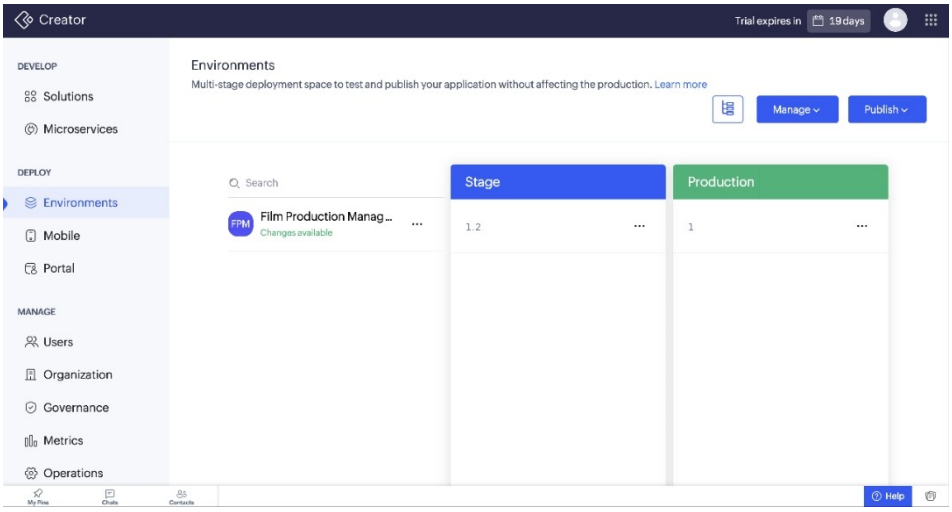


Figure 7.1: Environments – Film Production Management System Ready to Publish to Stage and Production

Table 7.1 is the checklist that is followed before each release. Items that are not yet done are shown with a cross so that the status of the project is clear.

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	Test cases designed and executed in Development (Table 6.1)	✓
2	Open defects corrected and re-tested (Table 6.2)	✓
3	Sample/test data removed or isolated	✗
4	Roles, permission sets and portal permission configured	✓
5	Users added and invitations accepted	✓
6	Blueprint and approval enabled	✓
7	Application published to Stage and verified	✓
8	Application published to Production and URL verified	✓
9	Version and release notes recorded	✓

All items except item 3 are complete: the test cases were run, the open defects were corrected and re-tested, the role hierarchy, the six permission sets and the Film_Portal_User permission exist, the users were added, both the Movie Production Blueprint and the equipment approval are enabled, and the application has been published to Stage and Production. Item 3 is marked with a cross because the sample data has not yet been removed or isolated from live data.

7.2 Performance Optimisation

The application serves a single production house, so the data volume is small (ten films, ten equipment items and one dashboard) and response times were not a concern. Optimisation therefore concentrated on reducing the manual work and the duplicated data that the earlier process involved, and on keeping the logic small and reusable. The measures applied are listed in Table 7.2; the before and after values are counts of manual steps or records rather than timings.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Lookup fields instead of typed text	Shooting schedule form	Film, scene and location typed in every schedule	Chosen from the three master forms	Each name entered once; no spelling variants
Pivot report for equipment cost	pivot table report	Rental cost of ten items added by hand	One grand total of Rs.236,000 per day split by condition	Instant, always current totals
Dashboard summary cards	Home page	Several reports opened to find counts	Requests, budget, members and active projects on one page	One-click overview
Small named Deluge functions	Default namespace	Calculations repeated in forms	Five reusable functions	One place to change a rule
Mandatory fields and fixed formats	Forms	Free entry of dates, times and IDs	Date dd-MMM-yyyy, time HH:mm:ss, ID formats and * fields	Fewer incomplete or malformed records
Role-based menus	Sidebar and permission sets	Every user sees every form	Each role sees only its forms and reports	Smaller, faster-to-read screens

7.3 Security Features

Security in the application works at three levels: who may open a component (the role hierarchy and the permission sets), what an outside user may see (the portal permission) and how the account itself is administered (users, the super admin and the operations console). Because remuneration and budget data are commercially sensitive, the permission sets were designed so that they are not open to every user (non-functional requirement NFR2). The controls are summarised in Table 7.3.

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Role hierarchy (CEO, producer, director, production manager, crew member)	Settings → User Permissions → Roles	Users seeing records of people above them in the hierarchy
Six permission sets (Admin, Cinematographer, Crew Member, Director, Post-Production Manager, Producer)	Settings → User Permissions → Permissions	Unauthorised access to forms, reports and settings
Portal permission Film_Portal_User	Settings → Portal User Permissions	Actors and crew seeing more than their own films, schedules and tasks
Default Customer profile limited to add and view	Portal User Permissions	Outside users editing or deleting records
Super admin separated from invited users	Admin dashboard → Users	Changes to the application by non-administrators
Mandatory fields and field formats	All ten forms	Incomplete or malformed data entering the masters
Equipment approval	Equipments allocation form	Equipment being allocated without a decision
Operations console (Audit Trail, Logs, Backup, Domain Restriction, API Management)	Admin dashboard → Operations	Untraceable changes and loss of data

7.4 Governance and Administration

Logs and traceability. The Operations page of the Creator admin dashboard (Figure 7.2) groups the tools that govern an account: Backup, Audit Trail, Email Management, Domain Restriction, API Management, Blueprint Analytics, Logs, Publish, Form Email, Marketplace, Support Access, System Integrations, Zia, SMS Templates, Payment Gateways and Databridge. Audit Trail and Logs record who performed an action and when, and Blueprint Analytics reports on the stages through which films and schedules have passed.

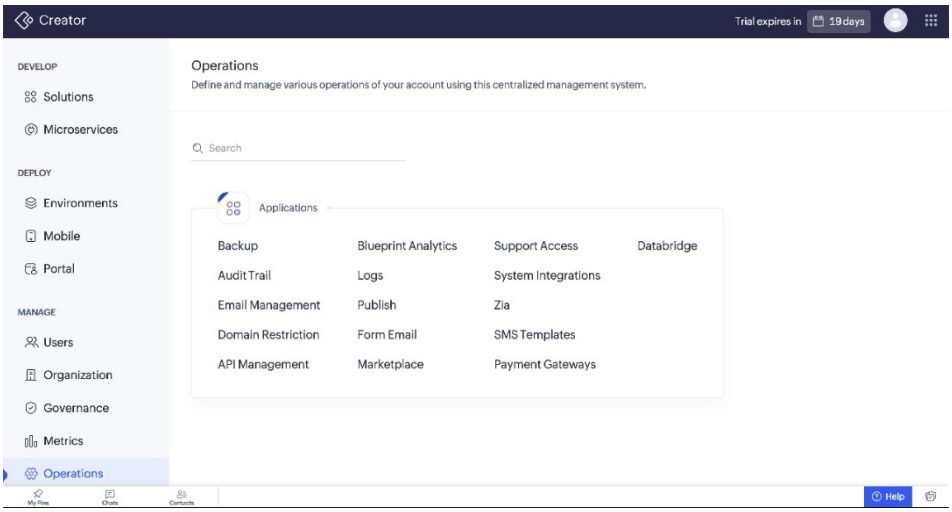


Figure 7.2: Operations Console of the Creator Admin Dashboard

Administration. Users are managed centrally on the Users page of the admin dashboard, where each user has a status and a last-activity entry. Menaha R is the super admin and the second user is invited by e-mail; roles and permission sets are then assigned inside the application. Naming conventions were followed throughout: forms carry the name of the master they hold, field link names match their labels (for example Person_ID), functions use descriptive names with underscores, and permission sets are named after the role they serve. Backup: the Backup tool of the Operations console and an exported copy of the application are kept after each milestone . The account is on a trial subscription that expires in about nineteen days, after which a paid plan is required to keep the application and its premium features such as Blueprints and approvals running.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The following scenario was used to demonstrate the application end to end. The producer registers a film in the Film Master form; the list report shows it at once with its genre, language, director, producer and start date (Figure 8.1). The production manager maintains the cast and crew, location, equipment and scene masters, and the director then opens the Shooting schedule form and chooses the film, scene and location from the lookup lists instead of typing them again (Figure 8.2). Equipment needed for the shoot is requested through the Equipments allocation form and goes to the equipment approval, while the pivot report shows the producer how much of the daily rental cost is tied to equipment in good condition, in need of repair or damaged.

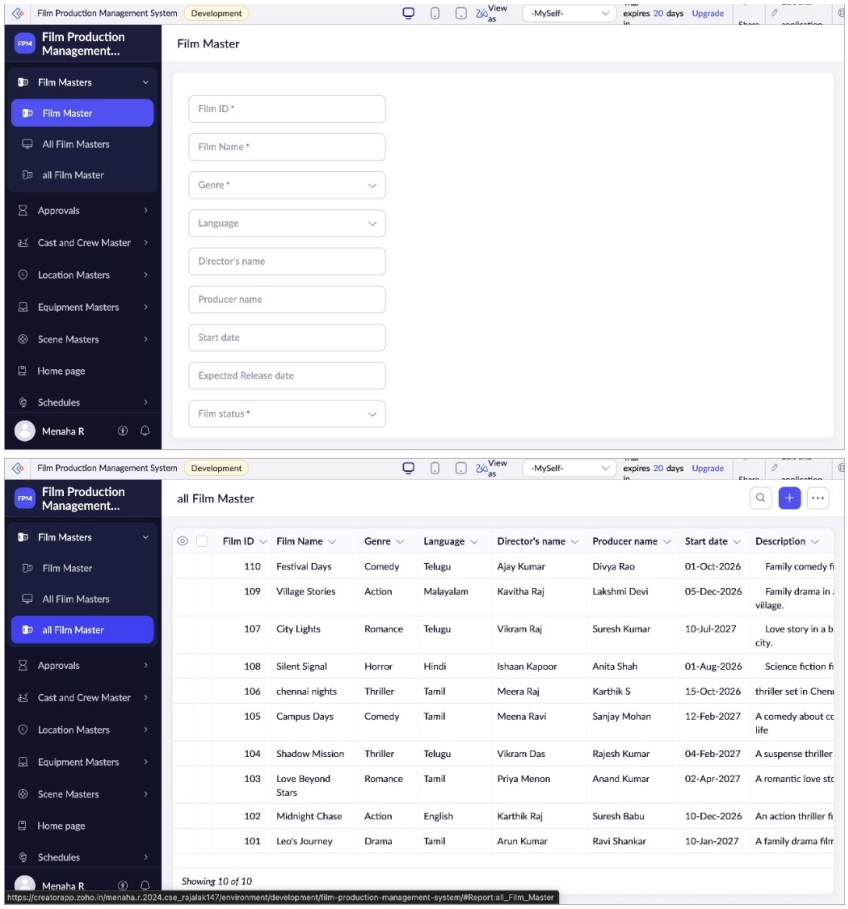


Figure 8.1: Scenario Step 1 – Film Master Form (top) and the All Film Master List Report (bottom)

On the shooting day the production manager records the attendance of each crew member as Present, Late, Half Day or Absent with a remark, and the attendance monitoring chart plots every crew member against the status (Figure 8.3, top). The remarks entered, such as ‘scene completed’ and ‘late pack-up’, appear inside the bubbles, so a director can see why a person was late or absent. The home page (Figure 8.3, bottom) then gives the producer a single overview of the 1,055 requests, the number of active projects (10), the total budget, the number of members and a doughnut chart of the equipment, and lets the producer search for a director by name.

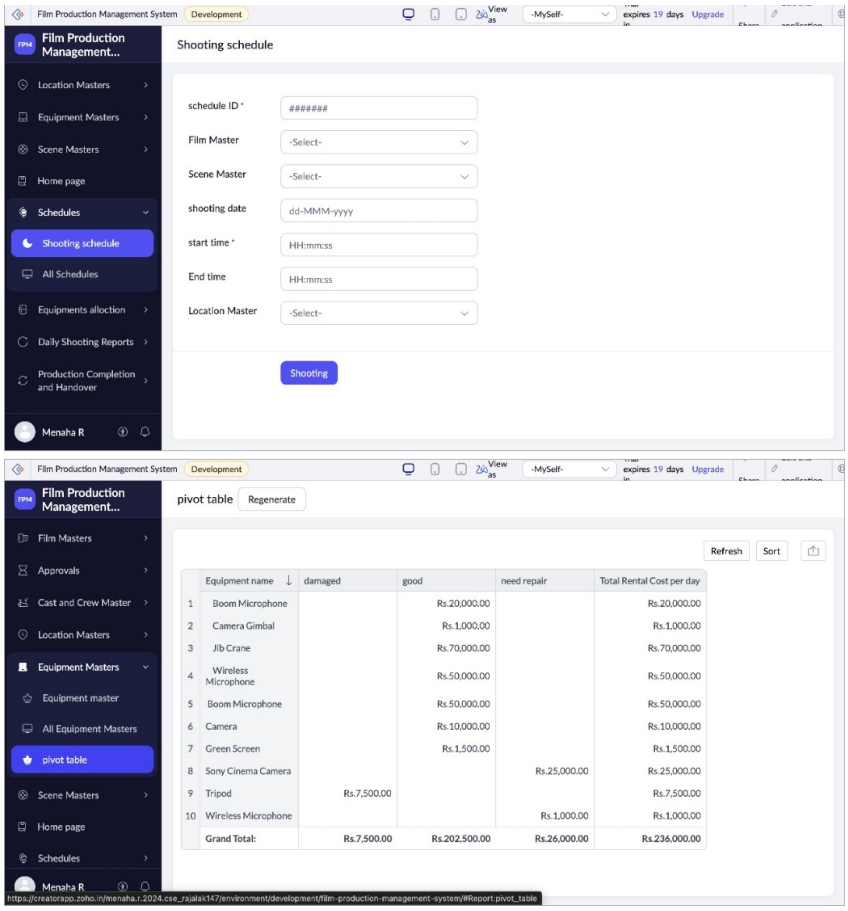


Figure 8.2: Scenario Step 2 – Shooting Schedule Form (top) and Equipment Pivot Report (bottom)

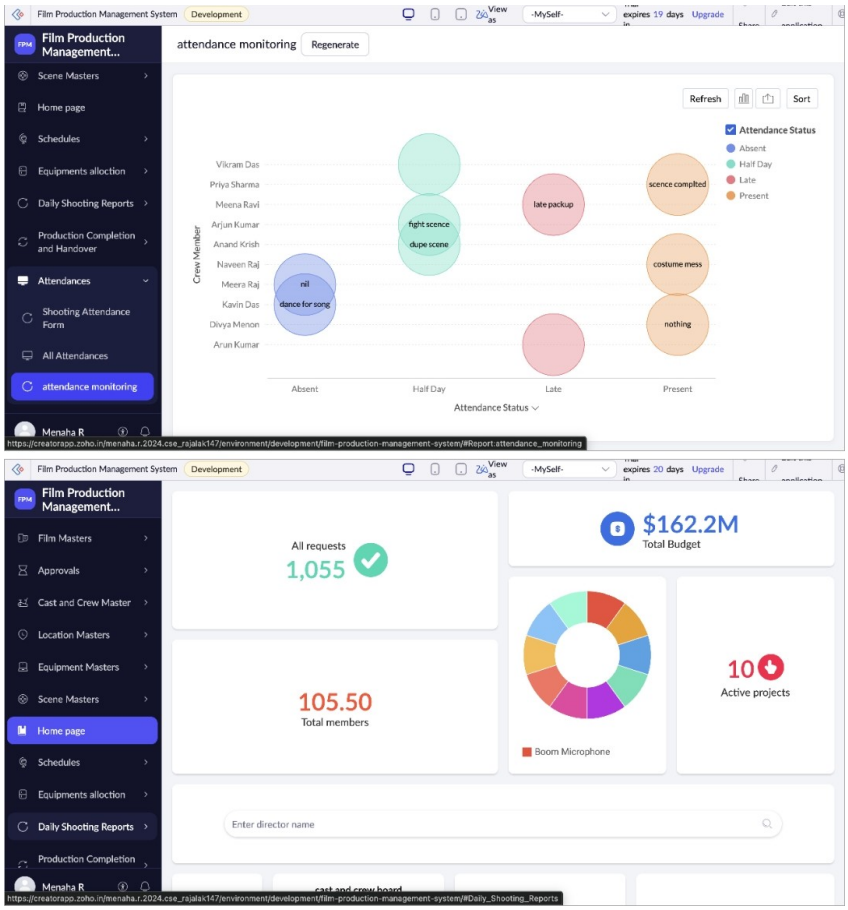


Figure 8.3: Scenario Step 3 – Attendance Monitoring Chart (top) and Home Page Dashboard (bottom)

Finally, the film is moved through its stages by the Movie Production Blueprint until the production completion and handover record is made. The same records feed every report, so no figure is entered twice.

8.2 Achievement of Objectives

Table 8.1 maps each objective of the project to the part of the application that meets it and to the evidence in this report.

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Maintain masters for films, cast and crew, locations, equipment and scenes	Sections 4.2, 4.3, 5.2	Ten forms with lookups; ten films and ten equipment items entered (Tables 4.2–4.6); TC-01 to TC-04
Plan shooting schedules without retyping data	Sections 5.2, 5.5	Shooting schedule with three lookups and mandatory fields; TC-05, TC-06
Control the use and cost of equipment	Sections 5.3, 5.6	Equipment approval and pivot report totalling Rs.236,000 per day; TC-07, TC-14
Record attendance and daily progress	Sections 5.2, 5.9	Attendance form, daily report form and monitoring chart; TC-08, TC-09, TC-15
Control the life cycle of a film	Section 5.6	Movie Production Blueprint on Film Master and Shooting schedule; TC-10
Give the producer reports and a dashboard	Sections 5.3, 5.4	Nine reports and one dashboard; TC-13 to TC-16; dashboard values to be corrected (BUG-02)
Restrict access by role and give a portal view	Sections 5.8, 5.11, 7.3	Role hierarchy, six permission sets and Film_Portal_User; TC-17 to TC-19
Deploy and govern the application	Chapter 7	Ready for Stage and Production; publishing pending (Table 7.1)

Seven of the eight objectives are met in the Development environment. The last, deployment, is complete in design but not yet in practice, because the application has not been published to Stage and Production.

8.3 Comparison with the Existing System

The existing process of Chapter 2 relied on registers, spreadsheets and messages. Table 8.2 compares it with the proposed application.

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Film and crew records	Separate sheets and registers	Master forms with lookups	One source of truth
Shooting schedules	Typed and circulated manually	Schedule form with film, scene and location lookups	No retyping; consistent names
Equipment tracking and cost	Cost added by hand	Equipment master, approval and pivot report	Instant totals by condition
Attendance	Paper or message based	Attendance form and monitoring chart	Status and remark per person, visible at once
Daily progress	Informal reports	Daily Shooting Report form	Recorded and searchable
Visibility for the producer	Asked from each department	Home page dashboard and reports	Overview on arrival
Access control	Open files shared by all	Role hierarchy and permission sets	Each role sees what it needs
Access on location	Needs the office file	Browser and Zoho Creator mobile app	Available anywhere

8.4 Limitations of the Low-Code Approach

Several limits appeared during the project. Reports and charts: the attendance monitoring bubble chart becomes crowded when several people share a status, and the home page cards depend on the field and the aggregate chosen, which led to the unusual values of BUG-02. Scripting: the Deluge function Calculate_Remaining_Budget uses fixed values, and reading live values needs fetch and aggregate statements that were not completed in this version (BUG-01).

Platform and cost: the application runs only on Zoho Creator, in the data centre of the account, and the trial subscription ends after the trial period, after which a paid plan is needed for features such as Blueprints, approvals and additional users. Data design: Creator does not stop the same equipment name being entered twice unless No duplicate values is set, which is why Boom Microphone and Wireless Microphone appear twice in the pivot (BUG-04). Each limitation has a workaround that is listed in Table 6.2 or in Section 9.2, and none prevented the core requirements from being met.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

No-code parts were built purely through configuration: the ten forms and their field properties, the lookups, the list, pivot, spreadsheet and bubble-chart reports, the home page dashboard, the role hierarchy, the six permission sets and the portal permission, the equipment approval and the Movie Production Blueprint. Low-code parts needed Deluge: the five custom functions for budget, shooting days, schedule check, movie category and rating validation. Pro-code was not needed: the Application IDE, widgets and variables were available but the standard components covered every requirement.

This distribution, with almost everything configured, a few business rules scripted and nothing hand-coded, shows where low-code is strongest: a small production house can obtain a working, secure application in a short time without a development team. The project would move towards pro-code if it needed a custom crew-facing interface, offline attendance on remote locations, or integration with accounting and payroll systems; at that point an external service would be more appropriate than more Deluge.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

This project built the Film Production Management System, a Zoho Creator application that brings the planning and running of a film into one place. The producer registers films and sees the budget, members and active projects on a dashboard. The production manager maintains the masters of cast and crew, locations, equipment and scenes. The director creates shooting schedules by choosing the film, scene and location from lookups, equipment is allocated through an approval, attendance and daily reports are recorded on location, and the Movie Production Blueprint moves a film through its stages. Role-based permission sets and a portal permission let each person see only what concerns them.

The project used the main capabilities of the platform covered in the course: data modelling, forms, reports, pages, Deluge functions, an approval, a Blueprint, role-based access and environments. It also showed where the tools reach their limits, such as the fixed values in the budget function, and these are recorded openly in the defect log. Fourteen of the twenty test cases passed, one failed and five remain to be completed.

9.2 Future Enhancements

- Correct the open defects of Table 6.2 and publish the application to Stage and then to Production.
- Make Calculate_Remaining_Budget read the budget from Film Master and the expenses from Daily Shooting Report, and create the weekly schedule that e-mails the production manager every Monday.
- Add an equipment ID and quantity with a duplicate check, and a scheduled reminder for items in the need repair condition.
- Add subforms for scene requirements and a call sheet that is generated and e-mailed to the cast and crew for each shooting day.
- Replace the bubble chart with a stacked bar chart of attendance by status and add a payroll report that multiplies attendance by daily remuneration.
- Connect the application to an accounting service such as Zoho Books for expenses, and to Zoho Cliq or e-mail for schedule notifications.

- Use Zia to suggest schedules that avoid clashes of crew, location and equipment.

9.3 Learning Outcomes and Career Relevance

The project developed skills across the low-code spectrum: modelling data with forms and lookups, designing reports and a dashboard, writing and executing Deluge functions, designing role-based access, building a Blueprint and an approval, and preparing a release through environments. It also showed the value of testing against requirements and of keeping an honest defect log.

These skills map to roles such as citizen developer, low-code developer, Zoho Creator or Zoho One solutions consultant and business-process automation engineer. The experience of finding where the platform and the AI assistant fall short is a step towards professional software development, which is where the no-code → low-code → pro-code spectrum of the course leads.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1 lists the Zoho Creator features expected in the course project and shows where each appears in this application. Y means fully implemented, P partially implemented and N not implemented. The features marked P or N are carried to Section 9.2.

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	Film Production Management System; sidebar grouped by master	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	Y	Number, Single line, Multi line, Dropdown, Date, Time, Phone, Email, Currency, Lookup	5.2	Tables 4.2–4.6
II	Lookup field (relationship between forms)	Y	Shooting schedule → Film Master, Scene Master, Location Master	4.2, 5.2	Table 4.1, Fig. 5.3
II	Subform (parent–child data)	N	No subforms used	4.2	—
II	Field validations / layout rules	P	Mandatory fields, number of digits, display formats	5.2	Fig. 5.2
II	List report with filter, grouping, custom action	P	All Film Masters, All Schedules, All Attendances (search, sort)	5.3	Table 5.1
II	Kanban / Calendar / Timeline / Map report	N	Not used	—	—
II	Pivot table / chart report	Y	pivot table; attendance monitoring bubble chart	5.3, 5.9	Fig. 5.4, 5.9
II	Page / dashboard built with UI Builder	Y	Home page	5.4	Fig. 5.5
II	Deluge – form events (validate / success / field actions)	P	Functions written; calls from form events	5.5	Table 5.2

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Deluge – functions, fetch, loops, collections	P	Five functions; fetch and loops not yet used	5.5	Code 5.1, C.1–C.5
II	Email / in-app notifications	P	Approval actions	5.6	Table 5.3
II	Form workflow	P		5.6	Table 5.3
II	Scheduled workflow	N	Not configured	—	—
II	Approval process	Y	equipment approval on Equipments allocation	5.6	Fig. 5.7
II	Blueprint	Y	Movie Production on Film Master and Shooting schedule	5.6	Fig. 5.7
II	Batch / report workflow	N	Not configured	—	—
II	Testing & debugging evidence	Y	20 test cases, 8 logged defects	6	Tables 6.1, 6.2
III	Integration with a Zoho service	N	None configured	5.7	Table 5.4
III	Connection / REST API call (invokeurl)	N	Not used	5.7	—
III	Webhook or Creator REST API	N	Not used	5.7	—
III	Users, roles and permission profiles	Y	Role hierarchy; six permission sets; two users	5.8	Fig. 5.8, Table 4.7
III	Record-level access criteria / customer portal	Y	Portal permission Film_Portal_User	5.11	Fig. 5.10
III	Zia / AI feature	Y	Zia Build Agent available in the builder; analytics in Section 5.9	2.5, 5.9	Fig. 5.9
III	Analytics (charts / Zoho Analytics)	Y	Dashboard cards, doughnut chart, pivot, bubble chart	5.4, 5.9	Fig. 5.5, 5.9
IV	Performance optimisation applied & measured	P	Design-level measures counted in steps	7.2	Table 7.2
IV	Deployment across environments	P	Development in use; Stage and Production pending	7.1	Fig. 7.1

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
IV	Security controls (permissions, portal)	Y	Role hierarchy, permission sets, portal permission	7.3	Table 7.3
IV	Governance (audit trail, administration)	P	Operations console available	7.4	Fig. 7.2
V	Pro-code extension (Catalyst / widgets / custom functions)	N	Not used; Application IDE and widgets available	5.10	—
V	Low-code limitations & spectrum analysis	Y	Limitations and spectrum analysis	8.4, 8.5	—
—	Mobile app / portal access	P	Portal permission; mobile screenshot to be added	5.11	Fig. 5.10

Total features: fully implemented 14 / 32, partially implemented 10 / 32, not implemented 8 / 32.

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	3 – High	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	3 – High	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	2 – Medium	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	2 – Medium	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	3 – High	

CO3 is self-assessed as Medium because roles, permission sets and the portal permission were configured, but no connection to an external service or REST API was used (Table A.1). CO4 is self-assessed as Medium because the roles, permissions and operations tools are in place, but the application has not yet been published to Stage and Production and the open defects of Table 6.2 remain.

APPENDIX C – COMPLETE DELUGE SOURCE CODE

This appendix reproduces the Deluge functions of the application. They are listed in the Default namespace under Workflow → Functions (Figure 5.6). The source of Calculate_Remaining_Budget is taken from the function editor; the source of the other four functions must be pasted from the editor before submission.

Table C.1: Deluge Functions in the Default Namespace

Function	Language	Returns	Purpose
Calculate_Remaining_Budget	Deluge	float	Returns the budget left after spending
Calculate_Shooting_Days	Deluge	int	Counts the number of shooting days
Check_Shooting_Schedule	Deluge	string	Checks a schedule and returns a message
Movie_Category	Deluge	string	Returns the category of a movie
Validate_Movie_Rating	Deluge	string	Validates the rating entered for a movie

C.1 Function – Calculate_Remaining_Budget

```
// Function : Calculate_Remaining_Budget
// Location : Workflow → Functions → Default
// Returns  : float

float Calculate_Remaining_Budget()
{
    total = 5000000;
    spent = 2000000;
    remaining = total - spent;
    return remaining;
}
```

The function returns 3,000,000. Total and spent are fixed in this version (BUG-01).

C.2 Function – Calculate_Shooting_Days

```
// Function : Calculate_Shooting_Days
// Location : Workflow → Functions → Default
// Returns  : int
// Purpose  : Counts the number of shooting days

// [paste the source code of this function from the function
editor here]
```

C.3 Function – Check_Shooting_Schedule

```
// Function : Check_Shooting_Schedule
// Location : Workflow → Functions → Default
// Returns  : string
// Purpose  : Checks a schedule and returns a message

// [paste the source code of this function from the function
editor here]
```

C.4 Function – Movie_Category

```
// Function : Movie_Category
// Location : Workflow → Functions → Default
// Returns  : string
// Purpose  : Returns the category of a movie

// [paste the source code of this function from the function
editor here]
```

C.5 Function – Validate_Movie_Rating

```
// Function : Validate_Movie_Rating
// Location : Workflow → Functions → Default
// Returns  : string
// Purpose  : Validates the rating entered for a movie

// [paste the source code of this function from the function
editor here]
```

C.6 Blueprint and Approval Configuration

The Blueprint and the approval are configured in the builder and carry no hand-written Deluge in the screens captured. Their settings are recorded in Table C.2.

Table C.2: Workflow Configuration Records

Name	Type	Applies to	Settings
Movie Production	Blueprint (enabled)	Film Master, Shooting schedule	Created 30-Oct-2026; stages and transitions
equipment approval	Approval (enabled)	Equipments allocation	Triggered on review; four actions; created 30-Oct-2026

APPENDIX D – USER MANUAL

D.1 Accessing the Application

1. Accept the invitation e-mail from Zoho Creator and sign in with the same e-mail address.
2. On the web, open creator.zoho.in, select the account menaha.r.2024.cse_rajalak147 if the account switcher shows another account, and open Film Production Management System.
3. On a phone, install the Zoho Creator app, sign in with the same account and open Film Production Management System.
4. Actors and crew members without a team role sign in through the portal and see only the Film_Portal_User view.

D.2 Instructions for the Producer and the Production Manager

- Register a film: open Film Masters → Film Master, fill Film ID, Film Name, Genre, Language, director, producer and dates, choose the Film status and submit. The film appears in All Film Masters.
- Maintain masters: add people in Cast and Crew Master (role, mobile number, experience, daily remuneration, available status and joining date), and add locations, equipment with its condition and rental cost, and scenes in their own groups.
- Review the project: open Home page to see requests, total budget, members, active projects and the equipment chart; type a director's name in the search box to find the films.
- Allocate equipment: open Equipments allocation, choose the film and equipment and submit; the request goes to the equipment approval.
- Check cost and condition: open Equipment Masters → pivot table to see the daily rental cost by condition.
- Close the film: move the film through its stages with the Movie Production Blueprint and complete the Production Completion and Handover form.

D.3 Instructions for the Director, Crew Members and Portal Users

- Plan a shoot (Director): open Schedules → Shooting schedule, enter the schedule ID, choose the film, scene and location, set the shooting date, start time and end time, and submit with the Shooting button.
- Record attendance (Production Manager or Crew Member): open Attendances → Shooting Attendance Form, choose the crew member and the status (Present, Late, Half Day or Absent), add a remark and submit; view the result in All Attendances and attendance monitoring.
- Report the day: open Daily Shooting Reports and fill the report for the shooting day.
- View your own work: crew members open their assigned tasks and shooting schedules; portal users open their film details, schedules and tasks through the portal.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	MENAHA R	Register No.	240701308
Project Title	Film Production Management System	Date	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner Name: _____	Signature of External Examiner Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	MENAHA R	Register No.	240701308
Department / Section	Computer Science and Engineering	Date of Viva	
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER Name: _____	EXTERNAL EXAMINER Name: _____